Université Paris Cité

A Statistical Analysis of Vowel Inventories of World Languages

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Instructions — please read carefully — I mean really carefully

- **Deadline: 30 September 2026, 8:00 a.m.** Reports handed in late will not be marked.
- Perfect is the enemy of good: it is better to answer fewer questions properly than to rush through the whole assignment and do a poor job of everything.
- **What to send, by email:** a PDF whose name is your family name, prefixed with `m2_lab1_` (so, for me, and for me only: `m2_lab1_wisniewski.pdf`), together with a `zip` archive containing your source code.
- **For each question**, the report must give (a) a summary of the question and (b) your answer, written out (code, results, analysis). Keep it *as short as possible*: what is being marked is that you understood what you did and that you can interpret the result. There is no need for 300 lines of code: include only the essential or tricky part. As I said (with the greatest conviction) last year, most of the questions you will meet (in my labs and, more generally, in your life) can be answered in three lines of Python.
- **The report must open with a short text of 500 words, plus or minus 10 %** (that is, between 450 and 550 words — you can thank me for having done the maths for you), describing (a) what you did, (b) the difficulties you ran into, (c) the answers you arrived at and (d) what you learnt (most important part).
- Evaluation criteria: correctness of the answers, depth of analysis, code quality, originality, quality of reasoning, self-critical thinking, and quality of writing.
- **Sensible use of AI is allowed** (you will admire the artistic vagueness surrounding the word "sensible"), on two conditions: you document how you used it, and you explain how you made sure that what it produced was correct.

1 Introduction

The aim of this practical session is to demonstrate how data-analysis methods and linguistic databases can be used to compare the characteristics of languages and to "discover" linguistic knowledge. This is a highly speculative lab: I have no idea what the answers to most of the

questions are (indeed, most of them are open-ended). The intention is rather to provide you with a guide to a set of experiments in the hope that (i) you will discover plenty of exciting things (ideally while enjoying yourselves), (ii) you will acquire useful knowledge and skills, in this case, in data analysis, and (iii) you will look back fondly on your younger years, when extraordinary teachers taught you the basics of pandas and statistics.

One of the aims of this practical session is to examine two well-known proposed linguistic universals relating to vowel systems. First, that virtually all languages possess the basic vowel triangle `[i]`, `[a]`, `[u]` (or close equivalents). Second, that if a language has highly marked vowels (for example nasal, long, or front rounded vowels), it almost always also has the corresponding "simpler" vowels. These claims are frequently cited in typological and phonetic studies (Maddieson 1984; Crothers 1978).[1] In this lab, we will test whether these are indeed universals by drawing on `PHOIBLE`, an open, cross-linguistic database of phonological inventories covering 2,185 languages (Moran and McCloy 2019). `PHOIBLE` aggregates data from multiple published sources and encodes segmental inventories in a standardised format, making it a valuable resource for large-scale comparisons of vowel systems across languages.

2 Getting the Data

The data are distributed as a single table. Download the version tagged `v2.0`:

`https://raw.githubusercontent.com/phoible/dev/v2.0/data/phoible.csv`

Do *not* use the `master` branch: it is a moving target, and it differs from `v2.0` both in its columns and in its sources — which is already a first lesson about working with data published on the web. All the figures quoted below refer to `v2.0`.

Each row of the file describes *one phoneme of one inventory*. The columns you will need are `InventoryID`, `Glottocode` and `ISO6393` (language identifiers), `LanguageName`, `Phoneme`, `SegmentClass` (which takes the values `consonant`, `vowel` and `tone`), `Marginal` (`TRUE`, `FALSE` or `NA`) and `Source` (the published description the inventory comes from); the remaining 37 columns encode phonological features and may be useful later.

1. Have you read and understood the instructions?

2. Load the file and report three numbers: how many rows it contains, how many distinct inventories (column `InventoryID`), and how many distinct languages (column `Glottocode`). Why are these three numbers different?

3. Why are there several ISO standards for identifying languages (and what are they)? Why did linguists also introduce Glottocodes? Is that a clever idea?

[1] For the braver among you, Lindblom and Maddieson (1988) apply the same kind of reasoning to consonant systems.

4. In `v2.0`, 532 languages have more than one inventory, because several linguists have described the same language and, of course, they do not always agree. Before computing anything else, you need one inventory per language. For this lab, adopt the following convention: for each `Glottocode`, keep only the inventory with the highest `InventoryID` (i.e. treat the most recently added description as the reference one), and discard the others. Do you think this is a reasonable choice? What alternatives could you have used instead (a randomly chosen inventory, a fixed preference for a given source, merging the inventories), and what would each of them cost you? Keep in mind that every result in this lab depends on this decision.

5. Measure how much the descriptions disagree: for a few languages described several times, compare the inventories. Do they have the same size? The same phonemes?

6. Compare the distribution of inventory sizes across the values of `Source`. Do the sources agree? What would it mean for your conclusions if one source systematically reported larger inventories than another?

7. State explicitly what you will count as a vowel. For this lab, adopt the following criterion: a vowel is any row with `SegmentClass == "vowel"`, and you keep it regardless of its `Marginal` value (`TRUE`, `FALSE` or `NA`). Write this criterion down — you will need it in every later section — and discuss the advantages and disadvantages of this choice compared with the alternative of excluding all `Marginal == TRUE` phonemes, and with the alternative of excluding rows where `Marginal` is `NA`.

With these conventions fixed, we can now compute some basic descriptive statistics.

8. Using the inventory-selection and vowel-definition conventions you have just settled on, compute the average number of vowels per language and the distribution of vowel inventory sizes.

9. Explain why it is interesting to examine these statistics.

10. Identify outliers using the interquartile range (IQR) method. Briefly explain the method and comment on your findings.

3 Testing Linguistic Universals

The basic idea of this part is to use the data to test whether the proposed universals are actually observed across the sample of languages in the `PHOIBLE` database.

11. Identify the three most frequent vowels (those occurring in the largest number of languages).

12. Compute the proportion of languages that contain each vowel of the basic vowel triangle [i], [a], [u], and discuss whether this supports the first universal.

13. The first universal is stated for [i], [a], [u] "or close equivalents". Give an explicit criterion for deciding what counts as a close equivalent (does a nasalised, long or otherwise modified [i] count as an [i]?), and recompute the proportions under your criterion. Does the conclusion change? This is a good illustration of how much a typological result can depend on a decision that looks purely technical.

14. An *absolute* universal admits no exception; a *statistical* universal only a few. Count the languages that violate the first universal and inspect a handful of them. Do the data support an absolute universal, a statistical tendency, or neither?

15. Are the exceptions genuine counter-examples, or artefacts of the description (transcription choices, phonemes marked as marginal, a single idiosyncratic source)? Check a few of them in the database itself.

3.1 The Second Universal: An Implication

The second universal has a different logical form: it does not state that some vowel is frequent, but that the presence of a marked vowel *implies* the presence of its simpler counterpart.

16. Build the list of languages having at least one nasal vowel, at least one long vowel, and at least one front rounded vowel ([y], [ø], ...).

17. Test the implication for each of these three cases: does a language with a nasal vowel also have the corresponding oral vowel? Does a language with a long vowel also have the corresponding short one? Does a language with [y] also have [i] and [u]? Report, in each case, the proportion of languages satisfying the implication and the number of exceptions.

18. An implicational universal "if A then B" corresponds to a specific pattern in a 2×2 table counting languages. Which cell of that table must be (nearly) empty? Why is this pattern *not* the same thing as a symmetric association between A and B?

19. Compare the two universals: which one is better supported by PHOIBLE? Which one is more informative about the structure of vowel systems?

3.2 Association Between Pairs of Phonemes

We will now investigate whether two phonemes tend to occur together in the same languages more often than would be expected by chance. For each pair (A, B) of phonemes:

20. Construct a 2×2 contingency table indicating, for each language in the dataset:

- whether it has phoneme *A*,
- whether it has phoneme *B*.

The four cells of the table will then correspond to: (*A* present & *B* present), (*A* present & *B* absent), (*A* absent & *B* present), (*A* absent & *B* absent).

21. Compute a χ^2 test of independence on this table. Does the result suggest that the two phonemes occur independently, or that there is a statistically significant association?

22. Compute the *mutual information* between the two phonemes, defined as

$$I(A;B) = \sum_{a\in\{0,1\}} \sum_{b\in\{0,1\}} p(a,b) \log \frac{p(a,b)}{p(a)\,p(b)}, \tag{1}$$

where $p(a,b)$ is the proportion of languages with the configuration (a,b), and $p(a)$ and $p(b)$ are the marginal probabilities of each phoneme. Interpret the value. Mutual information is always non-negative: a value close to zero indicates independence, but a high value does not tell you whether the two phonemes tend to co-occur or to exclude each other. How can you determine the direction of the association (e.g. by comparing $p(1,1)$ with $p(a{=}1)\,p(b{=}1)$)?

23. Compare the results of the two methods (χ^2 and mutual information). Do they lead to similar conclusions about the relationship between the two phonemes?

24. The χ^2 test is only valid if the expected count in each cell is large enough (a common rule of thumb is 5). Check this condition for a pair involving a rare phoneme. When it fails, use Fisher's exact test instead and compare the two p-values.

25. Suppose you now test *all* pairs of phonemes. How many tests does that represent? At the 5 % level, how many significant results would you expect by chance alone if no pair were associated? Explain, in your own words, why testing many pairs like this is a problem, and why a correction is needed. Then apply a correction for multiple testing (Bonferroni, or Benjamini–Hochberg) and report how many associations survive it.

26. With more than 2,000 languages, even a negligible association becomes statistically significant. Compute an effect size (ϕ coefficient or Cramér's V) for your pairs and rank them by effect size rather than by p-value. Is it the same ranking? Which of the two rankings is linguistically interesting?

27. Build a null model to calibrate your results: shuffle the inventories at random while keeping both the inventory sizes and the overall frequency of each phoneme unchanged, then recompute your statistics on the shuffled data. How many of your associations remain when compared against this baseline rather than against the independence assumption?

28. What can you conclude?

4 Are Languages Independent Observations?

Every test you have run so far treats each language as an independent observation. This is, however, false: languages are related to one another, either genealogically (they descend from a common ancestor) or areally (they are spoken by neighbouring communities and influence one another). A property shared by fifty Indo-European languages may be a single historical accident inherited fifty times, rather than fifty independent pieces of evidence for a universal. This difficulty is known in typology as *Galton's problem*, and it affects every result obtained in the previous sections.

29. Using the genealogical information available for the languages of the sample (language family, and macroarea), determine which families and which areas are over-represented in `PHOIBLE`. Produce a table or a figure.

30. Draw a genealogically balanced sample — for example one language per family, chosen at random — and recompute the proportions of the previous section on this sample. Do your conclusions hold?

31. Repeat this sampling a large number of times and report the distribution of the proportions you obtain. How stable are your conclusions, and how would you report them honestly?

32. In the light of these results, what exactly does a statement such as "X % of the world's languages have `[i]`" mean?

5 Discovering Regularities in the Corpus

The idea now is to see whether it is possible to identify other regularities from the data contained in the corpus. To do this, we will begin by looking at whether there are any *frequent itemsets*.

As a historical aside: frequent-itemset mining is one of the oldest problems in data mining, and one with substantial applications (read: applications that make a great deal of money). You are warmly encouraged to spend a few minutes reading about this family of methods, if only to round out your culture as budding data scientists.

A frequent itemset is a set of items (in our case, features or phonemes) that occur together in the data more often than a given minimum threshold. In the context of language inventories, an item could be the presence of a particular phoneme, and an itemset would be a combination of phonemes appearing in the same language. For example, the pair {`[i]`, `[u]`} would form an itemset; if this pair appears in many languages, it can be considered a frequent itemset.

Frequent itemsets are evaluated using support, which measures how often the itemset appears relative to the total number of observations. Formally, the support of an itemset X is defined as:

$$\text{support}(X) = \frac{\text{number of records (languages) containing } X}{\text{total number of records (languages)}} \tag{2}$$

Only itemsets whose support exceeds a chosen *minimum support threshold* (e.g. 10% of all languages) are considered "frequent". This approach is commonly used in association-rule mining (such as the `Apriori` or `FP-Growth` algorithms) to uncover systematic co-occurrence patterns in large datasets.

Once frequent itemsets have been identified, it is often useful to go further and examine **association rules** between items. Two key measures used for this purpose are *confidence* and *lift*.

Confidence measures how often items in Y appear in transactions (or here, in languages) that already contain X. It reflects the conditional probability of Y given X. Formally, for a rule $X \to Y$:

$$\text{confidence}(X \to Y) = \frac{\text{support}(X \cup Y)}{\text{support}(X)} = P(Y|X)$$

A high confidence value means that when X occurs, Y also occurs in a large proportion of cases.

Lift compares the observed co-occurrence of X and Y with what would be expected if they were statistically independent. It tells us whether the presence of X increases (lift > 1), decreases (lift < 1), or has no effect on (lift = 1) the likelihood of Y. Formally:

$$\text{lift}(X \to Y) = \frac{\text{support}(X \cup Y)}{\text{support}(X) \times \text{support}(Y)}$$

Lift thus normalises confidence by the baseline frequency of Y.

In the context of phoneme inventories, high confidence for a rule such as {/i/} → {/u/} would mean that most languages containing /i/ also contain /u/. A lift greater than 1 would indicate that /i/ and /u/ co-occur more often than expected by chance.

Using the `PHOIBLE` dataset:

33. For each possible itemset of size 1 (single phonemes), compute its *support* (proportion of languages containing that phoneme).

34. For all itemsets of size 2 (pairs of phonemes), compute their *support* in the same way.

35. Define as **frequent itemsets** those whose support exceeds the *minimum support threshold* chosen for this practical (e.g. 10% of all languages).

36. Identify the frequent itemsets of size 2 and comment on any regularities you observe.

37. For at least one of the frequent itemsets you identified, compute the *confidence* and *lift* for the association rule $X \to Y$ where X and Y are individual phonemes within that itemset.

38. Interpret your results: which phoneme combinations are most typical across languages? Do the confidence and lift values indicate stronger or weaker associations than expected under independence?

To go beyond manually calculating support for pairs of phonemes, you can use an existing implementation of the **Apriori algorithm**, which efficiently discovers frequent itemsets of any size. One convenient implementation is provided in the Python library `mlxtend` (Machine Learning Extensions). Its `apriori` function takes a binary matrix (rows = languages, columns = presence/absence of phonemes) and returns all itemsets above a specified minimum support threshold. The same library also includes an `association_rules` function for computing confidence and lift automatically.

39. Prepare your dataset as a binary matrix indicating, for each language, whether each phoneme is present (1) or absent (0).

40. Use the `apriori` function to identify frequent itemsets of size 2 and, if you wish, larger itemsets by adjusting the minimum support threshold.

41. Export the resulting itemsets together with their support values.

42. Using the `association_rules` function, compute the confidence and lift for at least two association rules of your choice and comment on your findings.

43. Run the analysis on vowels only, and then on the whole inventory. The itemsets computed over all phonemes are dominated by a handful of very frequent consonants: explain why, and say which of the two analyses answers the question we are interested in.

44. Vary the minimum support threshold (for instance 5 %, 10 %, 25 %) and report how the set of frequent itemsets changes. How would you choose a threshold, and what does this sensitivity tell you about the method?

45. Find a rule whose confidence is high but whose lift is close to 1. Explain why confidence alone is misleading, and what lift adds.

46. Do the rules discovered by `Apriori` recover the universals you tested earlier? Do they suggest any regularity that was not part of your initial hypotheses — and how would you check whether such a regularity is real?

6 Wrapping Up

47. `PHOIBLE` contains the languages that someone has described and published, not a random sample of the world's languages. Which languages are likely to be missing, and in which direction could this bias the universals you have tested?

48. Go back to the two universals stated in the introduction. For each, say what you now believe, what evidence supports it, and what would be needed to settle the question properly.

49. More generally: what kind of observation would convince you that a proposed universal is wrong?

Remember, when writing up: your results must be reproducible. State which version of the data you used, and how you evaluated the choices set out in this lab (the inventory-selection convention, the vowel definition, your significance threshold). Negative or inconclusive results are perfectly acceptable, provided they are argued for.

References

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Maddieson, Ian (1984). *Patterns of Sounds*. Cambridge: Cambridge University Press.

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